

Literature Status: Nonlinear 2-Local *-Homomorphisms of C^* -Algebras

Run fa_banach_001 — lane 0

9 August 2026

Status: literature already answered (partial subcase). The *-homomorphism half of Problem 3.12 in arXiv:1312.6850 was explicitly answered by Cabello–Peralta, arXiv:1510.00915, Corollary 2.8. The plain non-* homomorphism half is not covered.

1 Original question

Antonio M. Peralta’s *A note on 2-local representations of C^* -algebras*, arXiv:1312.6850, later published in *Operators and Matrices* **9**(2) (2015), 343–358, poses the following as Problem 3.12 on arXiv PDF page 12 [1].

Source Problem 3.12. *Is every (not necessarily linear) 2-local (*-)homomorphism between C^* -algebras a (*-)homomorphism? Equivalently, is the linearity hypothesis in Theorem 3.9 superfluous?*

The notation simultaneously asks two questions: one for bounded algebra homomorphisms used as the two-point interpolants, and one for *-homomorphisms. Source Theorem 3.9 already proves the conclusion once the 2-local map itself is linear (and bounded in the non-* case).

2 Separate later answer to the *-half

Juan Carlos Cabello and Antonio M. Peralta’s separate later paper, *Weak-2-local symmetric maps on C^* -algebras*, arXiv:1510.00915, published in *Linear Algebra and its Applications* **494** (2016), 32–43, proves automatic linearity for a larger symmetric class [2]. It then specializes this result on arXiv PDF page 8:

Later result.

*Every 2-local *-homomorphism between C^* -algebras is linear and is a *-homomorphism.*

The citation trail is exact. Corollary 2.7 gives linearity. The paragraph immediately before Corollary 2.8 combines that corollary with Theorem 3.9 of the source paper, cited there as reference [27], and says this yields the positive general result. Thus the later authors explicitly use the source theorem to remove its linearity hypothesis for 2-local *-homomorphisms.

3 Scope and search status

This settles the *-preserving half of Problem 3.12 for arbitrary source and target C^* -algebras. It does not settle the plain homomorphism half: the automatic-linearity mechanism in the

supporting paper applies to symmetric maps, and an arbitrary homomorphism need not preserve involution.

A bounded search through 9 August 2026 checked the run indexes, the exact wording of Problem 3.12, arXiv searches for nonlinear 2-local homomorphisms between C^* -algebras, and the source paper's citation trail. The decisive match was arXiv:1510.00915. Searches also recovered the earlier von-Neumann/compact- C^* result arXiv:1404.7597, but no general result for the plain non- $*$ half. No new mathematical result is claimed here.

References

- [1] A. M. Peralta, *A note on 2-local representations of C^* -algebras*, Operators and Matrices **9**(2) (2015), 343–358; arXiv:1312.6850. doi:10.7153/oam-09-20.
- [2] J. C. Cabello and A. M. Peralta, *Weak-2-local symmetric maps on C^* -algebras*, Linear Algebra Appl. **494** (2016), 32–43; arXiv:1510.00915. doi:10.1016/j.laa.2015.12.024.